

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Explain in detail the various types of unit cells with neat diagram. (CO2)
- Q.24 Describe TTT diagram in detail. (CO4)
- Q.25 Explain briefly the effect of various alloying elements of steel. (CO7)

No. of Printed Pages : 4

222022

Roll No.

2nd Year / Tool & Die Making **Subject : Material Science & Heat Treatment**

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 The ability of material to deform without breaking is called (CO1)
- a) Resistance b) Creep
- c) Plasticity d) Fatigue
- Q.2 The full form of B.C.C. is (CO2)
- a) Bio Centred Cubic
- b) Body Centred Cubic
- c) Bulge Centred Cubic
- d) Boron Centred Cubic
- Q.3 A thermal equilibrium diagram is also known as: (CO3)
- a) Phase diagram
- b) Constitutional diagram
- c) Both 'a' & 'b'
- d) None of these

- Q.4 Hypo-eutectic cast iron contains carbon _____
4.3% (CO4)
- a) Equal to b) More than
c) Less than d) None of these
- Q.5 Hooks law holds good up to (CO5)
- a) Limit of proportionality
b) Elastic limit
c) Yield point
d) Breaking point
- Q.6 18-4-1 HSS contains (CO7)
- a) 18% Tungsten b) 4% Chromium
c) 1% Vanadium d) All of these

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Define Ductility. (CO1)
- Q.8 Define Miller Indices. (CO2)
- Q.9 Define packing factor. (CO3)
- Q.10 Write the full form of T.T.T. diagram. (CO4)
- Q.11 Define Normalising. (CO6)
- Q.12 Define Non-Ferrous metals. (CO7)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Name various types of deformation processes, explain anyone with suitable diagram. (CO1)
- Q.14 Give classification of imperfections in crystalline material. (CO2)
- Q.15 Draw labeled diagram of Stress-Strain curve for Steel. (CO3)
- Q.16 Draw labelled Fe-C phase diagram. (CO4)
- Q.17 Write any five properties of ceramic materials. (CO5)
- Q.18 Give classification of composite materials. (CO5)
- Q.19 Explain in brief, the concept of case hardening. (CO6)
- Q.20 Enlist various heat treatment processes. Explain anyone. (CO6)
- Q.21 Describe cast iron and its types. (CO7)
- Q.22 Write any four remedies to prevent heat treatment defects. (CO7)